

Q.3 Answer the following questions.(Any Three)**[12]**

- [A] What is data independence? Explain the differences between physical and logical data independence.
- [B] Explain cardinality of relationship.
- [C] Explain the basic operations of relational algebra with example.
- [D] What is a key? Explain different types of keys with examples.
- [E] Consider the relation schemas as follows.
works(person name, company name, salary);
lives(person name, street, city);
located in(company name, city);
managers(person name, manager name);
1. Write a relational algebra query to List the names of the persons who work for company 'FBC' along with the cities they live in.
 2. Find the names of the persons who live and work in the same city.

SECTION II**Q.4 [A] Do as Directed.****[07]**

1. To remove a relation from an SQL database, we use the _____ command.
2. Which data manipulation command is used to combine the records from one or more tables?
 - a) Select
 - b) Project
 - c) Join
 - d) Product
3. SQL permits attribute names to be repeated in the same relation. **(True / False)**
4. Which operator is used to compare a value to a specified list of values?
 - a) Between
 - b) Any
 - c) In
 - d) All
5. A SQL query automatically eliminates duplicates. **(True / False)**
6. What is dual table in oracle?
7. DDL stands for _____.

Q.4 [B] Explain the following functions with suitable example.**[04]**

- a) To_char()
- b) To_date()
- c) Nvl()
- d) Months_between()

Q.5 Answer the following questions. (Any two)**[12]**

- [A] What is Normalization? Explain normalization upto 3NF using an example.
- [B] Write the syntax to Create a table, primary key and foreign key with suitable example.
- [C] Explain the SQL operators BETWEEN-AND, IN, LIKE and IS NULL by taking suitable examples.
- [D] Write the syntax of Inner join, Left Outer Join and Right Outer Join with suitable example.

Q.6 Answer the following questions. (Any three)**[12]**

- [A] Explain different types of anomalies.
- [B] Write the syntax of Update and Delete Commands.
- [C] What is a Subquery? Explain usage of IN, Exists, Any & All operators in subquery.
- [D] Explain *Group By* and *Having* clauses.
- [E] Explain the differences between DROP, DELETE and TRUNCATE commands.